

EFY Note

The source code of this project is available for free download at source.efymag.com

onrush of current from damaging the LEDs. Adding a 470-ohm resistor between data pins of Arduino and NeoPixel LEDs can help prevent spikes on the data line, which can damage the first LED.

Software

Circuit operation is controlled by the software program loaded into the internal memory of Arduino Uno. The program (main1.ino) is written in Arduino programming language. Arduino IDE is used to compile and upload the program.

ATmega328P on Arduino Uno board comes with a pre-programmed bootloader that allows you to upload a new code to it without using an

external hardware programmer. Connect Arduino board to the PC and select the correct COM port in Arduino IDE.

Along with Arduino program/sketch (main1.ino), you also need a header file (FastLED.h) for implementing multiple lighting effects and animation. Some major functions and operations used in the program code are:

#define LED_TYPE WS2812. It is an inbuilt function of FastLED.h to declare specific type of pixel LED packs like WS2811, WS2812, WS2812b, and WS2813.

#define COLOR_ORDER GRB. In this function you need to specify the order of the colour sequence of green, red, and blue colours.

#define BRIGHTNESS 64. In this function you can vary the brightness of LEDs, whose range is from 0 to 255 levels.

#define NUM_LEDS 50. In this function you specify the number of LEDs used (50 in this project).

Header file is included in FastLed-

master.zip file. Before compiling the sketch, you need to include FastLed-master.zip in the Arduino library.

Testing

After making the connections as shown in Fig. 5, upload the source code to Arduino Uno board. Connect Arduino board to 5V DC source through its USB cable by connecting to your PC/laptop. Then connect a 5V, 5A DC adaptor to the NeoPixel LED strip. You will see different LED light effects and patterns on the LED strip.

Note. While connecting a NeoPixel LED strip to any live-power source through an AC/DC adaptor, first connect the ground (-) terminal before connecting any other pin. Conversely, disconnect the ground terminal last when separating strip from the power source. **EFY**



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